Rodent Social Behavior Recognition using a Global Context-aware Vision Transformer Network

Muhammad Imran Sharif [1] Doina Caragea [1]

[1]Kansas State University, Department of Computer Science

Problem Statement

- Manual annotation of rat behavior is time-consuming and error-prone [1].
- Inconsistent annotations lead to unreliable behavioral data.
- Need for an automated deep learning system to recognize behaviors accurately.

Objectives

- Design a deep learning framework to recognize social behaviors in rodents.
- Leverage Global Context Vision Transformer (GC-ViT) to improve accuracy [2].
- Evaluate performance across behaviors using the RatSI dataset [3].

Proposed Method

We propose a Vision Transformer for Rat Social Interaction (ViT-RSI), inspired by GC-ViT [2].

- Uses Depthwise Separable Convolution-based Mixed-Scale Feedforward Network (DSC-MSFN) [4].
- Employs global tokens and multi-stage architecture for contextual modeling.

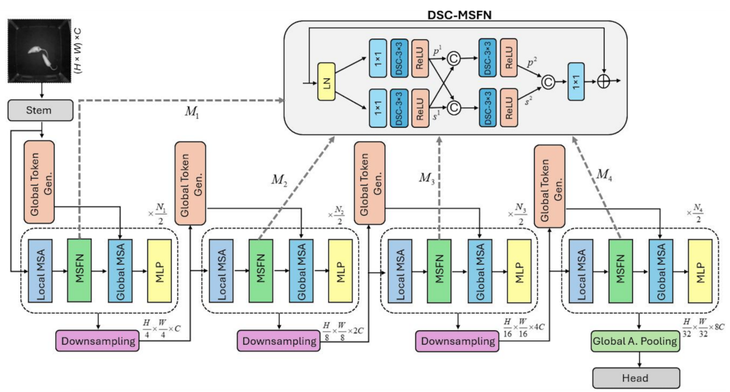


Figure 1. ViT-RSI architecture showing query-token interaction and context modeling across stages.

Evaluation Metrics

(i) $\text{Precision} = \frac{TP}{TP + FP}$

(ii) $\text{Recall} = \frac{TP}{TP + FN}$

(iii) $\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Dataset Description

RatSI dataset [3]:

- 5 behaviors: Approaching, Following, Nose Contact, Moving Away, Solitary.
- 9 videos at 30 FPS: 2 test, 7 train/validation.
- 260K+ labeled grayscale frames (224x224 px).

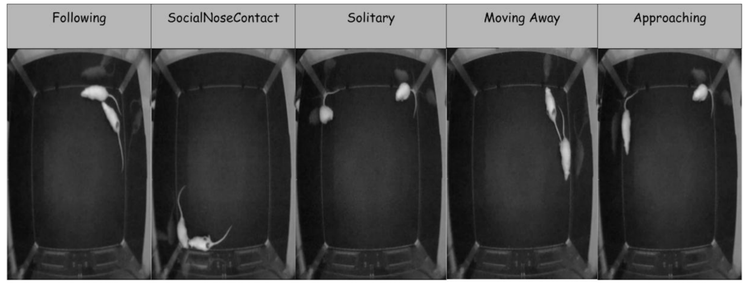


Figure 2. Sample frames showing annotated social behaviors.

Performance Summary

- **Solitary** class: highest F1-score (0.94).
- **Moving Away** and **Following**: F1 values are 0.86 and 0.81, respectively
- **Nose Contact**: Lowest (F1 = 0.48) due to class imbalance.

Action	Precision	Recall	F1 Score
Approaching	0.80	0.82	0.81
Nose Contact	0.80	0.34	0.48
Following	0.82	0.81	0.81
Moving Away	0.83	0.89	0.86
Solitary	0.93	0.95	0.94

Table 1. Action-wise performance scores for each social behavior.

Model	Method	Approaching	Contact	Following	Moving Away	Solitary
Lorbach et al. [3]	GMM	0.43	**0.58**	0.53	0.26	0.80
ViT-RSI (Ours)	ViT	**0.81**	0.48	**0.81**	**0.86**	**0.94**

Table 2. Comparison of F1 scores between ViT-RSI and Lorbach et al. [3] on RatSI behaviors.

Performance Metrics (F1 Score)

- "Solitary" and "Moving Away" showed highest classification accuracy.
- "Nose Contact" remains the most challenging behavior.

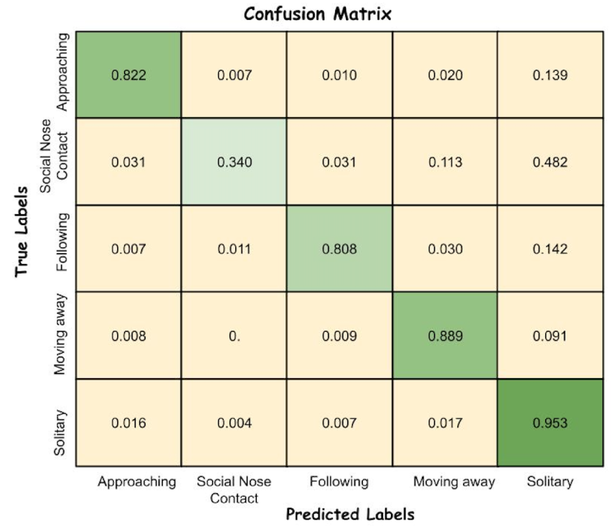


Figure 3. Confusion matrix showing model predictions.

Conclusion

ViT-RSI outperforms GMM-based approaches with better accuracy across most behaviors. Fewer parameters and better context modeling make it effective.

Future Work:

- Use frame sequences for temporal modeling.
- Apply semi-supervised learning to reduce annotation effort.

References

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